QMEE Project report

Mechanisms of acclimation to thermal and saline stress in the starlet sea anemone, *Nematostella vectensis*

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Introduction

The objective of this study was to determine how temperature and salinity acclimation responses interact to shape metabolic phenotype in the starlet sea anemone, *Nematostella vectensis*.

1.1 Hypothesis/Predictions

Thermal acclimation alters the metabolic phenotype of *N. vectensis*. The consequences of this shift will further depend on salinity. Warm temperatures should acutely increase metabolic rate by accelerating biochemical and mitochondrial processes, while high salinity further increases the metabolic demand. The predictions, therefore, are as follows:

(1) Routine metabolic rate and enzyme activities will increase with acute test temperature,
(2) Cold-acclimated anemones will exhibit higher routine metabolic rates and enzyme activities than warm-acclimated anemones when measured at a common test temperature, consistent with compensatory thermal acclimation, and
(3) The magnitude of this acclimation response will differ across salinities, with acclimation effects expected to be strongest in the cold and high salinity-acclimated treatment.

Methods

2.1 Experimental Design

Anemones used in this study were collected from the coast of New Hampshire, USA. Different clones within eight genotypes were acclimated from this population in an orthogonal, two-factor design — warm and cold treatments at 16°C or 24°C, and high and low salinity treatments at 30 ppt or 15 ppt artificial seawater for a period of two weeks. These salinities were chosen because these anemones were found at a salinity of 20 ppt in their tide pools during collection.

To characterize potential acclimation responses to these stressor interactions, the effects of acclimation on routine metabolic rate (RMR), on an indicator of metabolic capacity, citrate synthase (CS), and on total ATPase activity, which is related to osmoregulatory capacity, were measured at both test temperatures. Furthermore, thiobarbituric acid-reactive substances (TBARS) concentrations and protein carbonyl content (PCC) were measured to quantify oxidative damage across the different treatments.

2.2 Statistical analysis

The dataset was analyzed with a multivariate linear mixed model using the lme4 R package (Bates, Mächler, Bolker, & Walker, 2015). Since the measured variables are not independent, a multivariate model helps address this concern. The first model grouped RMR, CS and ATPase activity together because they shared similar experimental designs and were more causal in nature. This allows for direct evaluation of acclimation effects. Moreover, TBARS and PCC were

grouped in a separate model to assess oxidative damage and examine the consequences of the acclimation responses. In both models, all variables were allometrically scaled using the mass of the anemone(s) used.

2.2.1 Model 1 – RMR, ATPase, CS

```
log_measure ~ 0 + var_name +
              var_name:(AT * AS * TT) +
              var_name:log_mass +
              (1 | 0 + genotype:var_name)
```

The response variables were included on the log scale. The fixed effects were as follows:

(1) The name of the variable measured (RMR/CS/ATPase)
(2) The interaction between the variable measured and the acclimation temperature, acclimation salinity, and test temperature
(3) The scaling component to the mass of tissue is used on the log scale.

A random effect was also included at the genotype level. All intercepts were suppressed.

2.2.2 Model 2 – TBARS and PCC

```
log_measure ~ 0 + var_name +
              var_name:(AT * AS) +
              var_name:log_mass +
              (1 | 0 + genotype:var_name)
```

The response variables were included on the log scale. The fixed effects were as follows:

(1) The name of the variable measured (TBARS/PCC)
(2) The interaction between the variable measured and the acclimation temperature and acclimation salinity
(3) The scaling component to the mass of tissue is used on the log scale.

A random effect was also included at the genotype level. All intercepts were suppressed.

2.2.3 Model analytics (same for models 1 and 2)

Diagnostic plots were made using the DHARMa package (Hartig, 2016). Estimated marginal means and contrasts with 95% confidence intervals were calculated (Lenth & Piaskowski, 2017), and plotted using ggplot2 (Wickham, 2016) (Figures 1-4). Default settings in the contrasts function for p-value adjustments using the false discovery rate method were used across the eight tests.

Although genotype was included in this model as a random effect to account for within-group variation, it did not appear to contribute to the observed variation in acclimation responses. This means that across all eight genotypes, there are no clear differences that would alter the anemones' acclimation responses to thermal and saline stress. This could be due to the small size of the current dataset, as there is insufficient data to conclude that genotype contributes variance. Using a reduced rank function on the random effect parameter, it would be interesting to look at within-genotype correlations as well, however, this dataset may not be best suited for it.

We did not include the day on which each response was measured in the final model. While there was some variance for this factor for RMR, both CS and ATPase measures were carried out on a single day each. It would be worthwhile to include the day as a random effect for RMR alone to make this model more informative.

We also noticed that removing CS as a response variable appeared to improve model diagnostics. There was little variance in the CS measured, which likely masked the overall variance in ATPase and RMR measures. Experimentally, CS typically reflects the maximal activity of mitochondria, but it is highly unlikely that all the CS measured is constantly being used to produce energy. This is evident from the mismatch in trends seen between RMR and CS, especially at the high salinity.

Results and discussion

Routine metabolic rate increased with acute test temperature across treatments, supporting the prediction that warming elevates metabolic demand. Evidence for compensatory thermal acclimation was partially supported, as cold-acclimated anemones generally exhibited higher routine metabolic rates than warm-acclimated individuals at 30 ppt. Consistent with the third prediction, acclimation responses were salinity-dependent. The clearest separation between acclimation groups occurred at 30 ppt, whereas responses at 15 ppt were more strongly driven by acute temperature exposure.

Enzyme activity measurements suggested mechanistic differences underlying these patterns. Warm-acclimated anemones exhibited higher citrate synthase activity, ATPase activity, and lipid peroxidation, indicating increased mitochondrial capacity, energetic demand, and potentially oxidative stress. Together, these results indicate that thermal acclimation alters the metabolic phenotype of *N. vectensis*, but the expression of this phenotype depends on salinity.

Combining the two models to examine acclimation responses and potential trade-offs in oxidative damage would provide a more complete picture of the acclimation response in anemones. The initial experimental design could have included a measure of RMR as a second scaling variable, so that each response could be expressed as a function of RMR. Overall, through this work, we were able to delineate some responses to both the acute and chronic effects of thermal and saline stress.

Figures

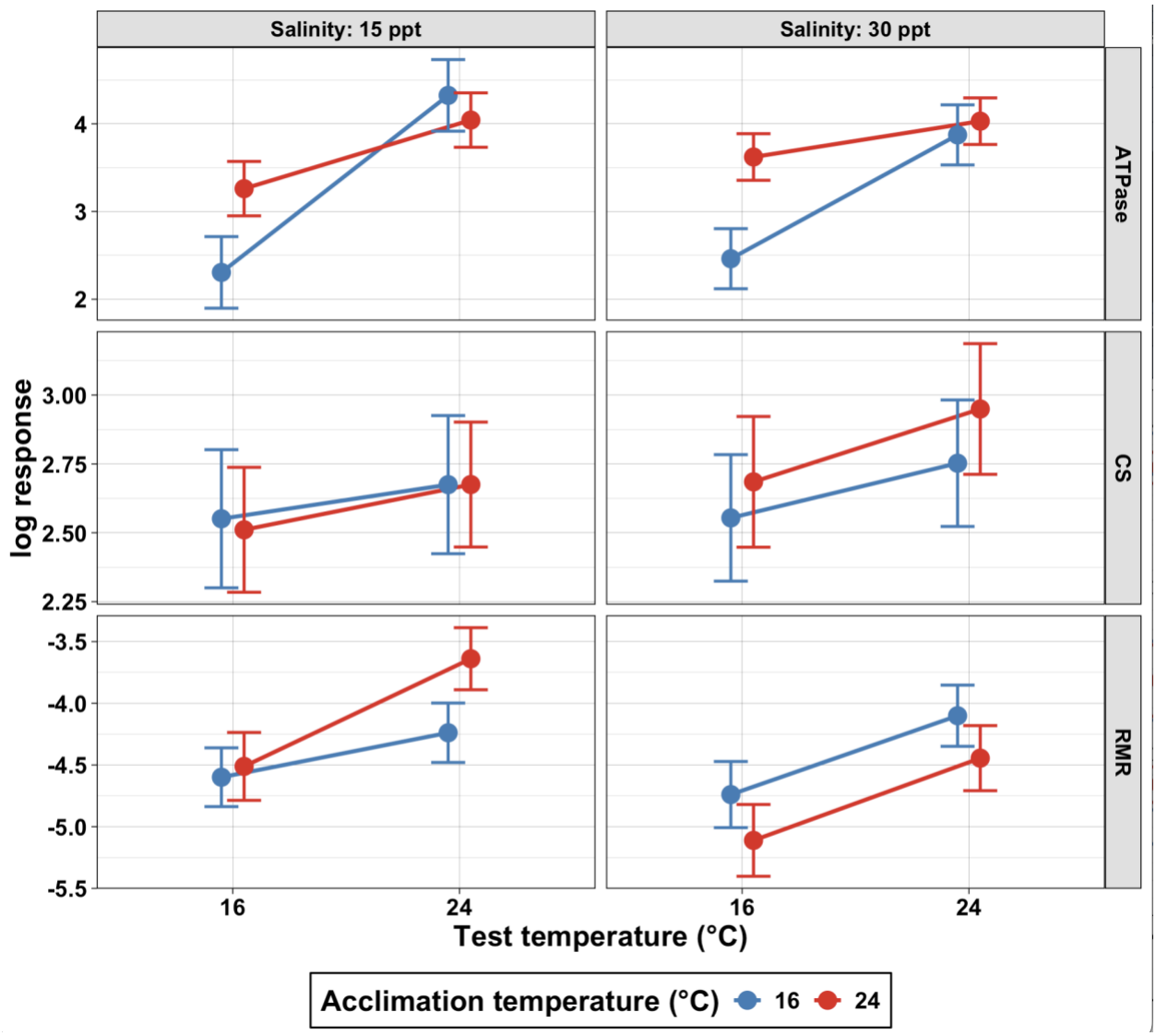

Figure 1: Estimated marginal means of measured ATPase activity, citrate synthase (CS) activity and routine metabolic rate (RMR) under different acclimation temperatures (16°C, 24°C), acclimation salinities (15 ppt, 30 ppt), and test temperatures (16°C, 24°C). Facets on the x-axis represent acclimation salinities, and facets on the y-axis show the name of the response variable measured. Test temperatures are shown on the x-axis. Error bars represent 95% CI.

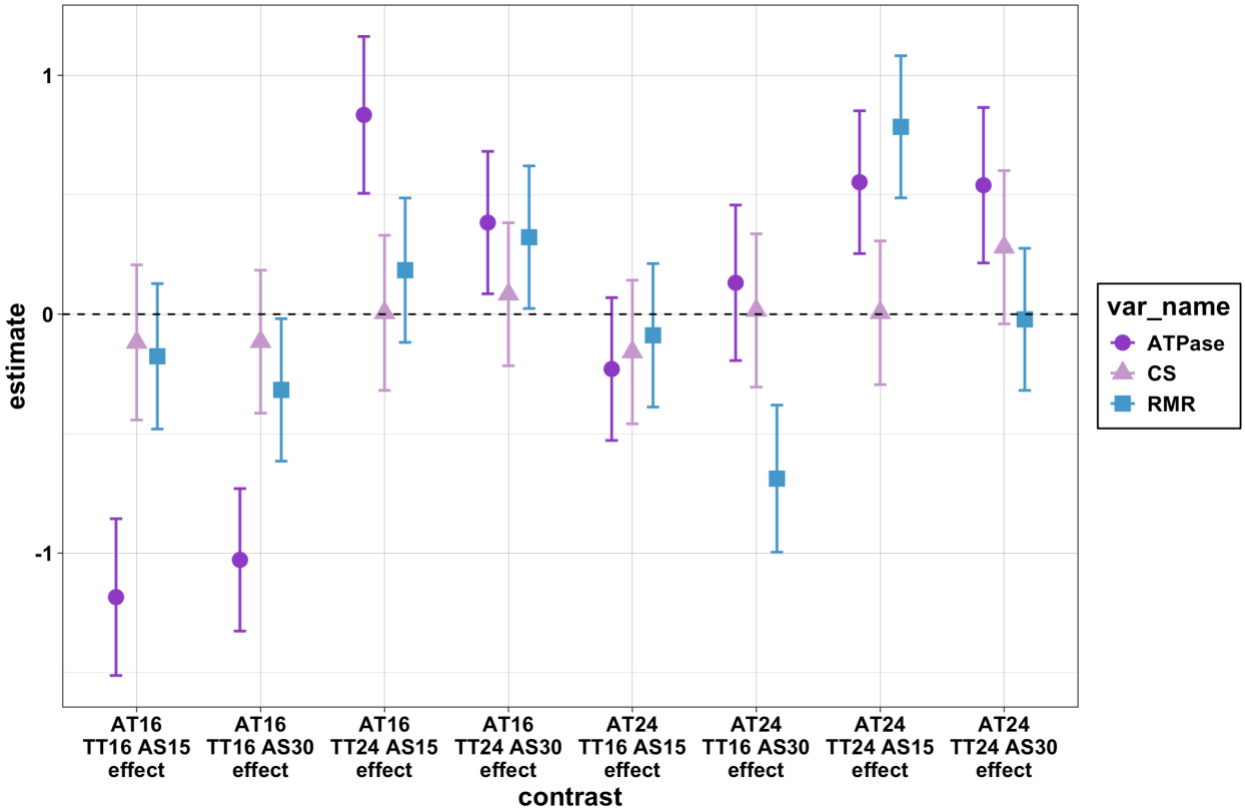

Figure 2: Custom contrasts to compare the effects of the different acclimation and test treatments of temperature and salinity on routine metabolic rate (RMR) as well as CS and ATPase activity. Error bars represent 95% CI. {AT: acclimation temperature (16°C, 24°C); AS: acclimation salinity (15 ppt, 30 ppt); TT: test temperature (16°C, 24°C)}

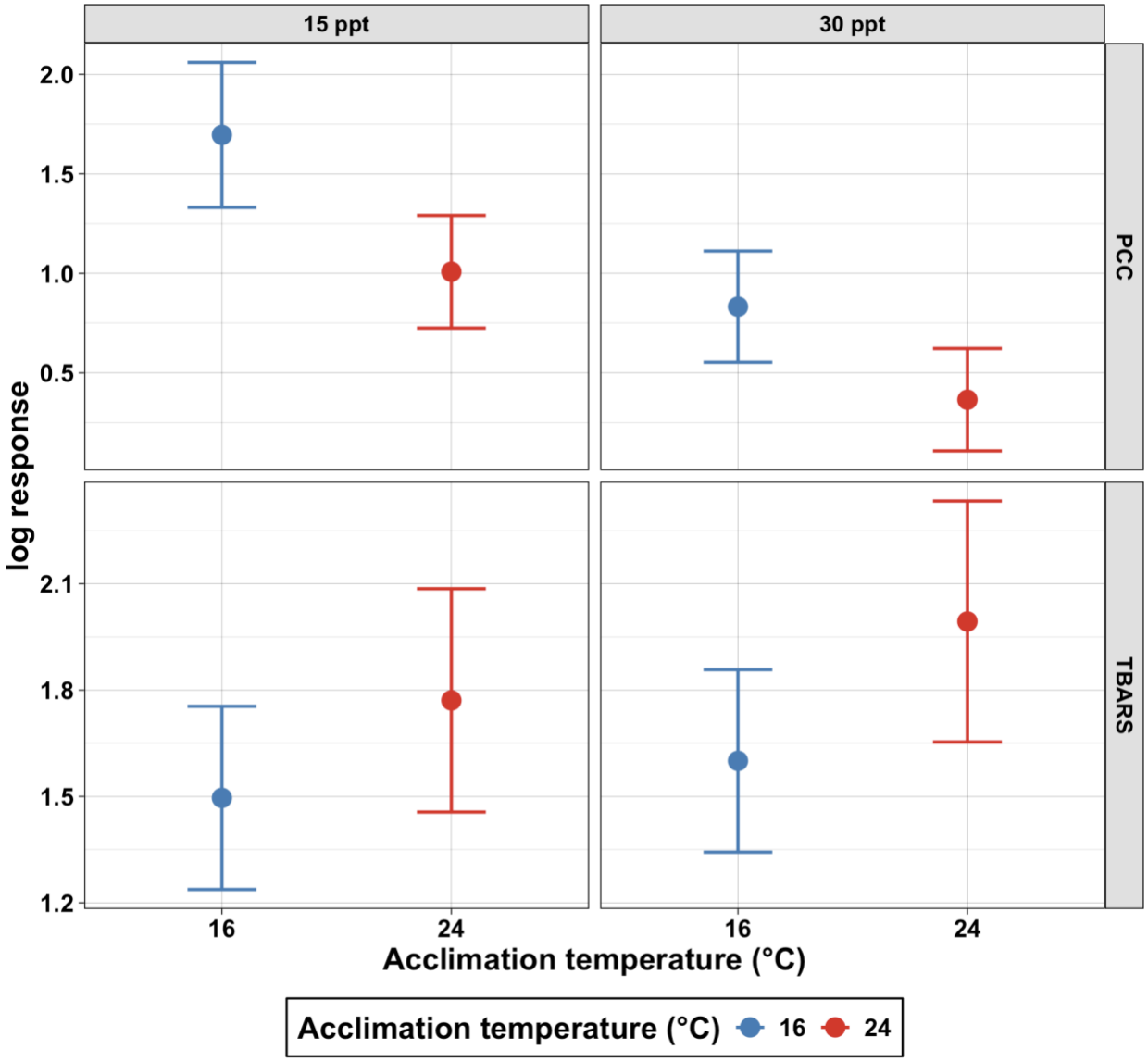

Figure 3: Estimated marginal means of measured PCC and TBARS under different acclimation temperatures (16°C, 24°C) and salinities (15 ppt, 30 ppt). Facets on the x-axis represent acclimation salinities, facets on the y-axis show the name of the response variable measured. Error bars represent 95% CI.

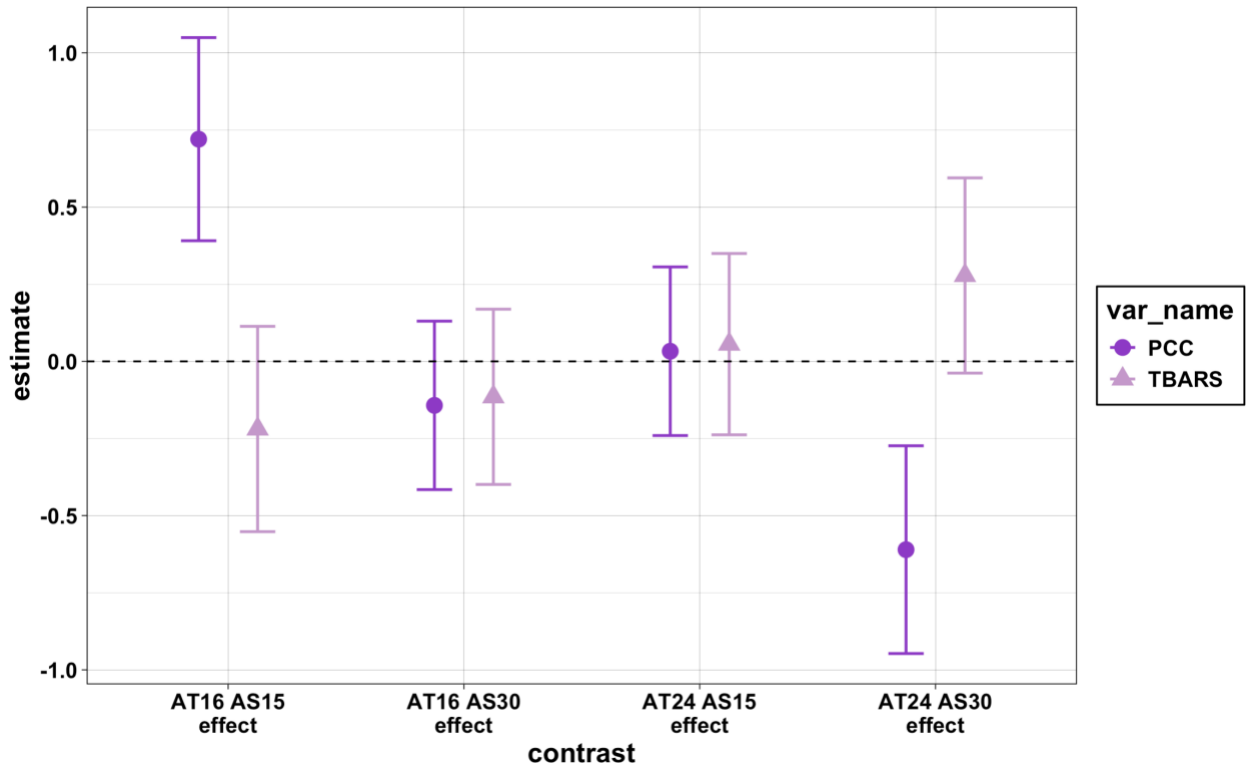

Figure 4: Custom contrasts to compare the effects of the different acclimation treatments of temperature and salinity on markers of oxidative damage as a result of acclimation (PCC and TBARS). Error bars represent 95% CI. {AT: acclimation temperature (16°C, 24°C); AS: acclimation salinity (15 ppt, 30 ppt)}

References

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, *67*(1). https://doi.org/10.18637/jss.v067.i01

Hartig, F. (2016). DHARMa: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models. *CRAN: Contributed Packages*. https://doi.org/10.32614/cran.package.dharma

Lenth, R. V., & Piaskowski, J. (2017). emmeans: Estimated Marginal Means, aka Least-Squares Means. *CRAN: Contributed Packages*. https://doi.org/10.32614/cran.package.emmeans

Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis | Springer Nature Link* (2nd ed.). Springer Cham. Retrieved from https://link.springer.com/book/10.1007/978-3-319-24277-4